

Turtle Environment Upgrade

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Element A

Problem Definition:

- The main problem with caring for turtles in captivity is creating an environment that supports their physical health, natural behaviors and long-term well-being as best as possible.

Why The Problem Exists:

- Captivity usually lacks conditions found in the wild, such as space, clean water systems, temperature, and access to UVB (UV B rays needed to produce Vitamin D3/absorb calcium) light for shell and bone health. Caretakers might also not have enough knowledge or resources to accurately recreate these conditions.

Three Possible Stakeholders:

1. Jorge Gomez- Jurado
 - a. Responsible for overseeing and maintaining turtles living conditions and health
2. Towson University (Biology department?)
 - a. Provides resources, space, and funding for proper care of turtles habitat
 - b. Hailey & Attia (Field Experts)
3. Animal welfare organizations
 - a. Concerned with ethical care and adhering to standards set in place for animals in captivity

Sources:

" Lethargy: A turtle lacking proper UVB exposure will typically show reduced activity levels. They may swim less, explore less, and generally appear less interested in their surroundings."

- <https://www.burapets.com/blogs/fish/uvb-lighting-for-turtles-explained-why-it-s-a-must-for-shell-health>

"UVB light (280–315 nm) is essential for turtles and tortoises because it enables the synthesis of vitamin D3 (cholecalciferol) in their skin. "

- https://ozarkexotics.net/education/f/the-critical-role-of-uv-light-in-turtle-and-tortoise-health?fbclid=IwY2xjawJYGY5leHRuA2FlbQlXMAABHVeL4GsgNaJTP2s9TFjrAsUU2TliZtYMMXCJmnySLuGKhspF0YSFDni-hQ_aem_TnQoLOmhJW2jvDfMPtLFHg&srs!tid=AfmBOoo1b32wMQEoo9YWAp4ofPaeHBdYKHHOBn0IT-ihK_xBtCs3u7UR

" The water in an aquatic turtle's enclosure should be maintained at 75 F with an aquatic heater....you'll need to use a submersible thermometer to ensure the tank's water stays within the ideal temperature range."

- <https://www.petmd.com/reptile/aquatic-turtle-care-sheet>

"Turtles need both land and water within their tank. As a rule, turtles need 10 gallons of water for 1 inch of shell. The water level also needs to be deep enough for your turtle to flip over and not get stuck – a depth of 1.5 times your turtle's length is ideal."

- <https://www.kingbritish.co.uk/help-advice/turtle-tank-setup-how-to#:~:text=Designing%20your%20turtle's%20environment,around%2025%25%20of%20the%20tank.>

"The dry landing place should be large enough to allow the turtle to climb completely out of the water. A large rock or a secure stack of smaller rocks with a flat top works well. The landing place should be easily accessible to the turtle and should be at the same end of the tank as the basking lamp."

- [https://vcahospitals.com/know-your-pet/turtles-aquatic-housing#:~:text=Aquatic%20turtles%20should%20be%20kept,82%C2%BAF%20\(24%C2%BAC%2D29%C2%BA C\).](https://vcahospitals.com/know-your-pet/turtles-aquatic-housing#:~:text=Aquatic%20turtles%20should%20be%20kept,82%C2%BAF%20(24%C2%BAC%2D29%C2%BA C).)

Pet. MD

Difficulty of Care	Intermediate
Average lifespan	20–40+ years with proper care, depending on species
Average Adult Size	5–12 inches long, depending on species
Diet	Juveniles are carnivorous; adults are omnivorous
Minimum Habitat Size	Varies depending on the turtle's body length; provide at least 10 gallons of tank space for every inch of body length; 40-gallon minimum recommended
PH	6.0–8.0